

CardFlow Credit Origination Platform

Multi-Tenant SaaS Solution for Credit Card
Application and Approval Automation

System Architecture Diagram

PREPARED BY: Jovi Sarmiento
CREATED ON: DD Month YYYY
(Updated on DD Month YYYY)
PREPARED FOR: [Client Name]
VERSION: Version XX

Table of Contents

1.	Introduction	1
1.1.	Purpose	1
1.2.	Scope	1
1.3.	Out of Scope	2
1.4.	Intended Audience	2
1.5.	Related Documents	2
2.	Architecture Overview	4
2.1.	Solution Architecture Overview	4
2.2.	Architectural Style	6
2.3.	Core Architectural Capabilities	7
2.4.	Architectural Principles	8
2.5.	Multi-Tenant SaaS Architecture Context	8
3.	Component Description	10
3.1.	Functional Components	10
3.2.	Component Interaction Overview	11
4.	Workflow Architecture	14
4.1.	Workflow Lifecycle Architecture	14
4.2.	Credit Application Processing Architecture	15
4.3.	Amendment Processing Architecture	16
4.4.	Reinstatement Processing Architecture	17
5.	Data Architecture	18
5.1.	Core Business Entities	18
5.2.	Entity Relationship Diagram	19
5.3.	Data Flow Architecture	20

6.	Security Architecture	21
6.1.	Authentication and Access Control	21
6.2.	Audit and Traceability	22
6.3.	Data Protection	23
7.	Integration Architecture	24
7.1.	External System Interfaces	24
7.2.	API Architecture	24
7.3.	Notification Architecture	27
8.	Deployment Architecture	29
8.1.	Environment Overview	29
8.2.	Deployment Diagram	29
8.3.	Scalability and Availability Considerations	30
9.	Architecture Considerations	31
9.1.	Scalability	31
9.2.	Maintainability	31
9.3.	Security	32
9.4.	Auditability	32
9.5.	Tenant Isolation	33
10.	Assumptions and Constraints	34
10.1.	Assumptions	34
10.2.	Constraints	34

List of Figures

Figure 1 Solution Architecture Overview.....	6
Figure 2 Multi-Tenant SaaS Architecture Context	9
Figure 3 Component Interaction Overview.....	13
Figure 4 Workflow Lifecycle Architecture	14
Figure 5 Credit Application Processing Architecture	15
Figure 6 Amendment Processing Architecture.....	16
Figure 7 Reinstatement Processing Architecture	17
Figure 8 Entity Relationship Diagram.....	19
Figure 9 Data Flow Architecture	20
Figure 10 Security Architecture.....	21
Figure 11 API Architecture.....	26
Figure 12 Notification Architectures	28
Figure 13 Deployment Diagram.....	30

List of Tables

Table 1 Primary Architectural Layer	4
Table 2 Architectural Principles	8
Table 3 Functional Components.....	10
Table 4 Component Interaction Overview	11
Table 5 Core Business Entities	18
Table 6 External System Interfaces	24
Table 7 API Architecture.....	26
Table 8 Notification Triggers.....	27
Table 9 Environment Overview	29
Table 10 Scalability and Availability Considerations	30
Table 11 Assumptions.....	34
Table 12 Constraints.....	35

Version History

[illegible]

1. Introduction

This section introduces the purpose, scope, and architectural context of the **Software Architecture Diagram Document** for the **CardFlow Credit Origination Platform**. It provides a high-level overview of the system architecture, key architectural components, integration points, and deployment considerations that support the platform's credit application processing workflows.

1.1. Purpose

The purpose of this **Software Architecture Diagram Document** is to present the architectural design of the **CardFlow Credit Origination Platform**, and provide a visual and conceptual representation of how the system components interact to support the end-to-end credit origination lifecycle.

This document aims to:

- Define the high-level architectural structure of the platform.
- Describe the major functional and technical components of the system.
- Illustrate workflow processing, data flow, and integration behavior.
- Present the relationships between application layers, services, and external systems.
- Support design review, development, testing, deployment, and maintenance activities.
- Provide architectural traceability across related project documentation artifacts.

1.2. Scope

This document covers the software architecture of the **CardFlow Credit Origination Platform**, including the major application components, workflow services, data management mechanisms, integration interfaces, and deployment structure required to support credit application processing within a multi-tenant SaaS environment.

The scope of this document includes:

- High-level solution architecture
- Functional and logical architecture views
- Workflow processing architecture
- Data flow and data management architecture

- Security and access control architecture
- Integration and API interaction architecture
- Deployment and environment architecture
- Architectural considerations related to scalability, maintainability, auditability, and tenant isolation

1.3. Out of Scope

Detailed implementation specifications such as database schema definitions, API endpoint specifications, and infrastructure configuration details are outside the scope of this document and are addressed in the **Technical Design Document (TDD)** and **API Documentation** artifacts.

1.4. Intended Audience

This document is intended for stakeholders involved in the design, development, implementation, testing, deployment, and maintenance of the **CardFlow Credit Origination Platform**, including:

- Solution Architects
- System Architects
- Software Developers
- Quality Assurance Engineers
- DevOps and Infrastructure Engineers
- Business Analysts
- Technical Writers
- Project Stakeholders

The document serves as a reference for understanding the architectural structure, system interactions, and deployment considerations of the platform throughout the system development lifecycle.

1.5. Related Documents

This **Software Architecture Diagram Document** is derived from and aligned with the following project artifacts:

- **Business Requirements Document (BRD)** – Defines the business objectives, operational requirements, and high-level business workflows
- **Software Requirements Specification (SRS)** – Defines the functional and non-functional system requirements
- **Functional Design Document (FDD)** – Defines workflow behavior, state transitions, routing logic, and functional processing rules
- **User Stories and Acceptance Criteria Document** – Defines user-centered functional scenarios and validation conditions

The architecture views and diagrams presented in this document reflect the workflows, functional behaviors, integration requirements, and system constraints defined across these related artifacts.

2. Architecture Overview

This section provides a high-level overview of the **CardFlow Credit Origination Platform** architecture, highlighting its solution structure, core capabilities, and design principles. The platform is designed to support configurable credit origination workflows while maintaining scalability, security, auditability, maintainability, and tenant isolation in a multi-tenant SaaS environment.

2.1. Solution Architecture Overview

The **CardFlow Credit Origination Platform** is a multi-tenant SaaS solution designed to automate and manage the end-to-end lifecycle of credit origination processes, including credit application submission, amendment processing, reinstatement processing, validation, review, routing, approval, and outcome management.

The platform provides a centralized environment where applicants, tenant users, reviewers, approvers, and administrative users interact with configurable workflow-driven services to process requests according to defined business rules and operational policies.

The solution architecture is structured around the following primary architectural domains.

Table 1 Primary Architectural Layer

Architectural Layer	Description
External Users	Users and stakeholders that interact with the platform to initiate, process, review, approve, and manage requests.
Presentation Layers	Provides web-based interfaces for applicants, tenant users, reviewers, approvers, and administrators.
Security and Access Control Layer	Enforces authentication, authorization, RBAC, tenant isolation, and secure access management.
Application Services Layer	Handles workflow orchestration, validation, routing, notifications, audit operations, and reporting services.

Architectural Layer	Description
Business Rules and Workflow Layer	Applies workflow logic, validation rules, eligibility checks, routing conditions, and state transition rules.
Data Management Layer	Manages application data, workflow records, documents, user information, and audit logs.
Integration Layer	Supports communication with external systems and third-party services.

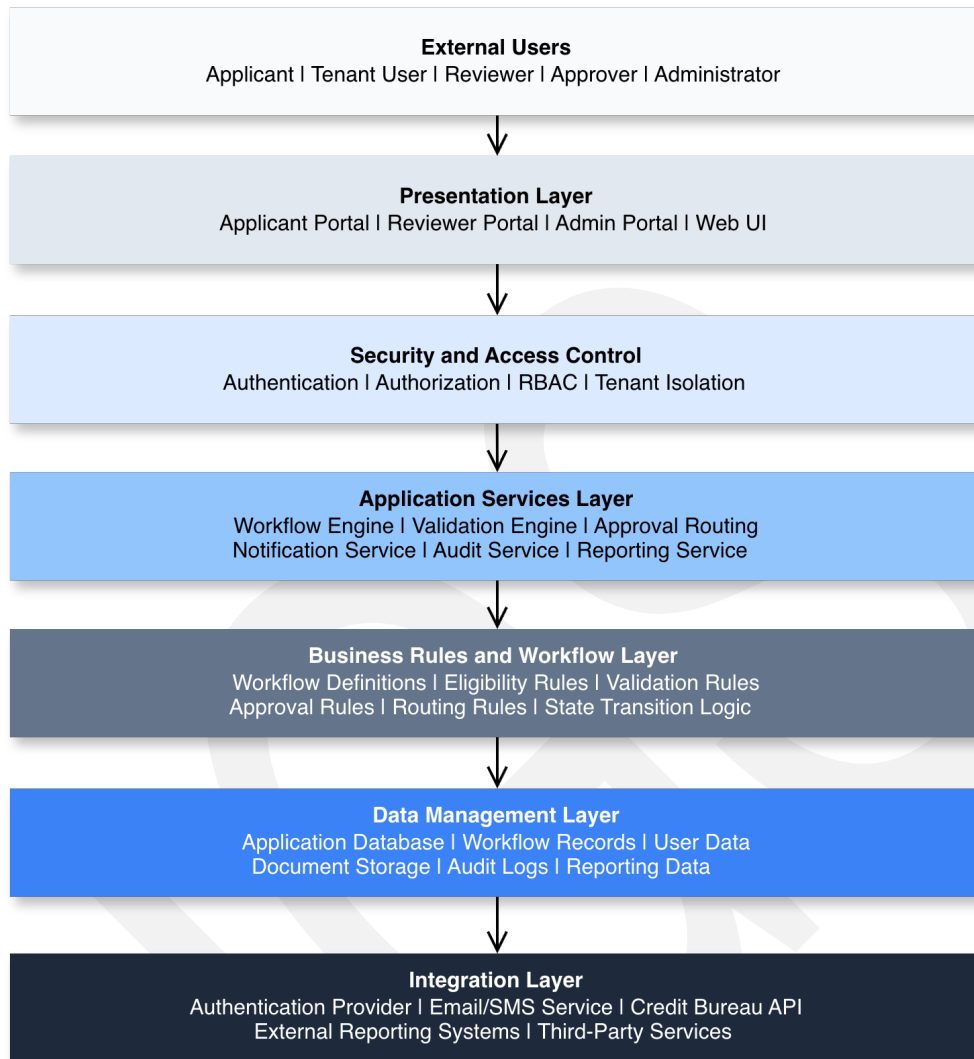


Figure 1 Solution Architecture Overview

The platform architecture supports configurable workflow processing to allow tenant organizations to adapt approval structures, routing behavior, validation logic, and operational rules based on business requirements.

2.2. Architectural Style

The **CardFlow Credit Origination Platform** adopts a layered service-oriented architecture designed to separate presentation, workflow processing, business logic, data management, and integration responsibilities into logically organized architectural components.

The architecture follows the following design principles:

- Separation of concerns across architectural layers

- Centralized workflow orchestration and lifecycle management
- Configurable business rule processing
- Role-based interaction and authorization
- Modular service interaction
- Tenant-aware data and configuration isolation
- Traceable and auditable workflow execution

The platform architecture is designed to support future extensibility by allowing additional workflow types, integrations, reporting capabilities, and operational configurations to be introduced without significantly impacting existing system components.

2.3. Core Architectural Capabilities

The architecture supports several core platform capabilities required for credit origination processing and workflow automation.

- **Workflow Management** – The platform provides centralized workflow orchestration for credit application, amendment, and reinstatement processing, supporting lifecycle transitions from initiation through outcome processing.
- **Validation and Eligibility Processing** – Validation services enforce mandatory field requirements, document requirements, eligibility conditions, and input constraints before requests are allowed to proceed within the workflow.
- **Review and Approval Processing** – The platform supports configurable review and approval workflows, including reviewer actions, approval routing, decision recording, returned requests, and rejection handling.
- **Document Management** – The architecture supports secure upload, association, retrieval, and management of supporting documents linked to workflow requests and applications.
- **Audit and Traceability** – Audit services record workflow actions, approval decisions, status transitions, validation outcomes, and user interactions to support operational traceability and compliance monitoring.
- **Multi-Tenant Configuration** – Tenant-specific configuration capabilities allow organizations to maintain independent workflow rules, approval routing behavior, operational settings, and user access structures within the shared SaaS environment.

2.4. Architectural Principles

The architectural design of the **CardFlow Credit Origination Platform** is guided by principles intended to support operational consistency, maintainability, scalability, and secure workflow processing.

Table 2 Architectural Principles

Principle	Description
Scalability	The architecture supports growth in transaction volume, tenants, and workflow processing activities.
Maintainability	Architectural components are logically separated to simplify updates and maintenance.
Configurability	Workflow behavior, routing logic, and operational rules can be configured per tenant.
Auditability	Workflow actions and decisions are recorded to support traceability and governance.
Security	Role-based access control and secure data handling mechanisms are enforced.
Reusability	Shared workflow services and validation components are reused across workflows.
Extensibility	Additional workflows and integrations can be introduced with minimal architectural impact.

2.5. Multi-Tenant SaaS Architecture Context

The **CardFlow Credit Origination Platform** operates in a multi-tenant SaaS architecture model where multiple tenant organizations share a common application environment while maintaining logical separation of data, workflow configurations, user access controls, and operational policies.

The architecture ensures that:

- Tenant-specific data remains logically isolated.

- Workflow configurations can vary per tenant organization.
- User access permissions are managed independently per tenant.
- Reporting and audit visibility are restricted according to tenant boundaries.
- Shared platform services can support multiple tenant environments simultaneously.

This architecture model enables centralized platform management while supporting configurable operational behavior across participating organizations.

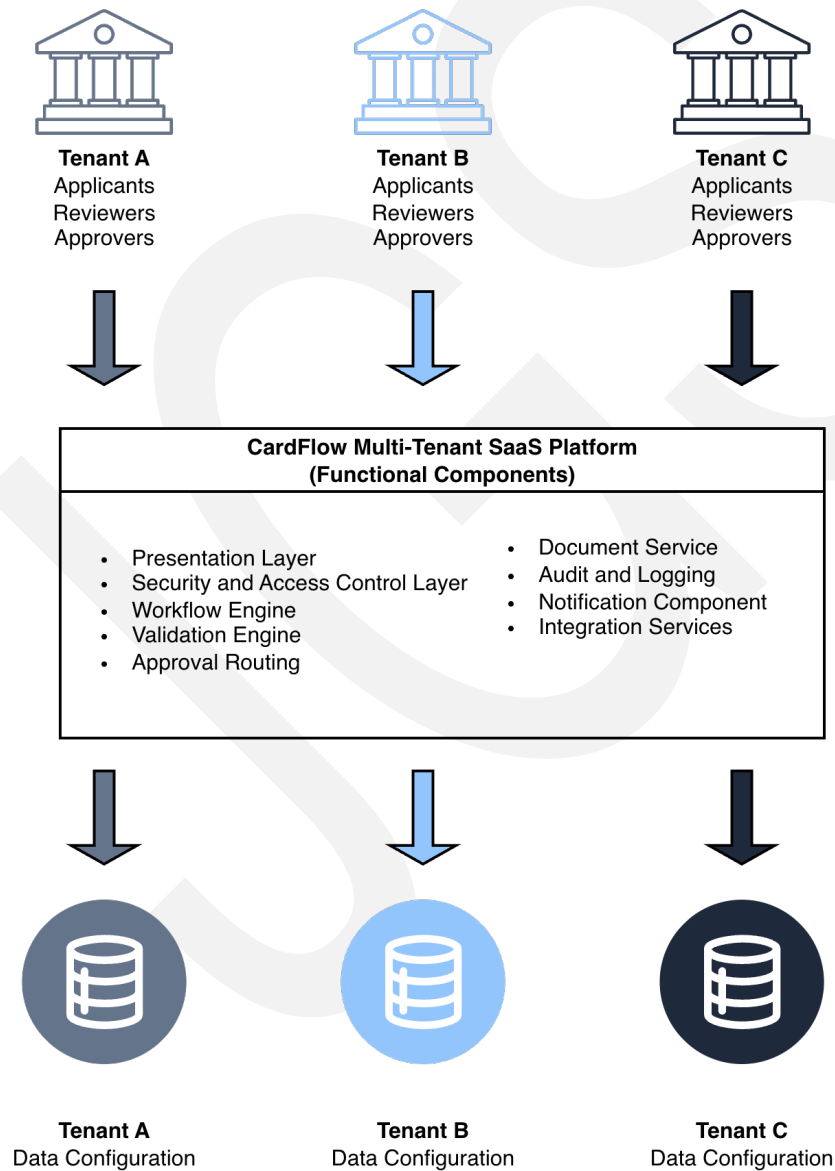


Figure 2 Multi-Tenant SaaS Architecture Context

3. Component Description

This section describes the major architectural components of the **CardFlow Credit Origination Platform** and their responsibilities in the overall system architecture. It provides a functional overview of the platform's core processing components and illustrates how these components interact to support workflow execution, validation, approval processing, data management, audit logging, and external integrations.

3.1. Functional Components

The **CardFlow Credit Origination Platform** is composed of multiple functional components that collectively support the end-to-end credit origination lifecycle across credit application, amendment, and reinstatement workflows.

Table 3 Functional Components

Functional Component	Description
Presentation Layer	Provides web-based interfaces for applicants, tenant users, reviewers, approvers, and administrators to interact with the platform.
Security and Access Control	Enforces authentication, authorization, role-based access control, and tenant isolation mechanisms.
Workflow Engine	Manages workflow orchestration, lifecycle transitions, routing logic, and workflow state progression.
Validation Engine	Performs mandatory field validation, eligibility checks, document validation, and business rule enforcement.
Approval Routing	Determines reviewer and approver routing based on configurable workflow and approval rules.
Document Service	Handles document upload, retrieval, association, and storage for workflow requests and applications.

Functional Component	Description
Audit and Logging (Reporting)	Records workflow actions, status changes, approval decisions, validation events, and system activities for traceability purposes. Supports operational reporting, application tracking, audit visibility, and workflow monitoring.
Notification Component	Generates and delivers workflow-triggered notifications through configured communication channels.
Integration Service	Facilitates communication with external systems and third-party services such as authentication providers, notification services, and external APIs.

3.2. Component Interaction Overview

This section illustrates how the major functional components of the **CardFlow Credit Origination Platform** collaborate to process user requests, enforce business rules, manage workflow progression, maintain audit records, and integrate with supporting services and external systems.

Table 4 Component Interaction Overview

Interaction Area	Description
User Request Processing	Users interact with the platform through web-based interfaces to initiate and manage workflow requests.
Validation Processing	Submitted requests are validated against workflow rules, eligibility conditions, and document requirements.
Workflow Orchestration	The workflow engine controls lifecycle progression, routing, and state transitions.

Interaction Area	Description
Approval Processing	Requests are routed to reviewers and approvers according to configurable approval rules.
Document Handling	Supporting documents are uploaded, stored, and linked to workflow requests.
Audit and Logging	Workflow actions and status changes are recorded for traceability and compliance purposes.
Notification Handling	Notifications are generated based on workflow events and approval outcomes.
External Integration	External services support authentication, notifications, and third-party integrations.

CardFlow Credit Origination Platform System Architecture Diagram

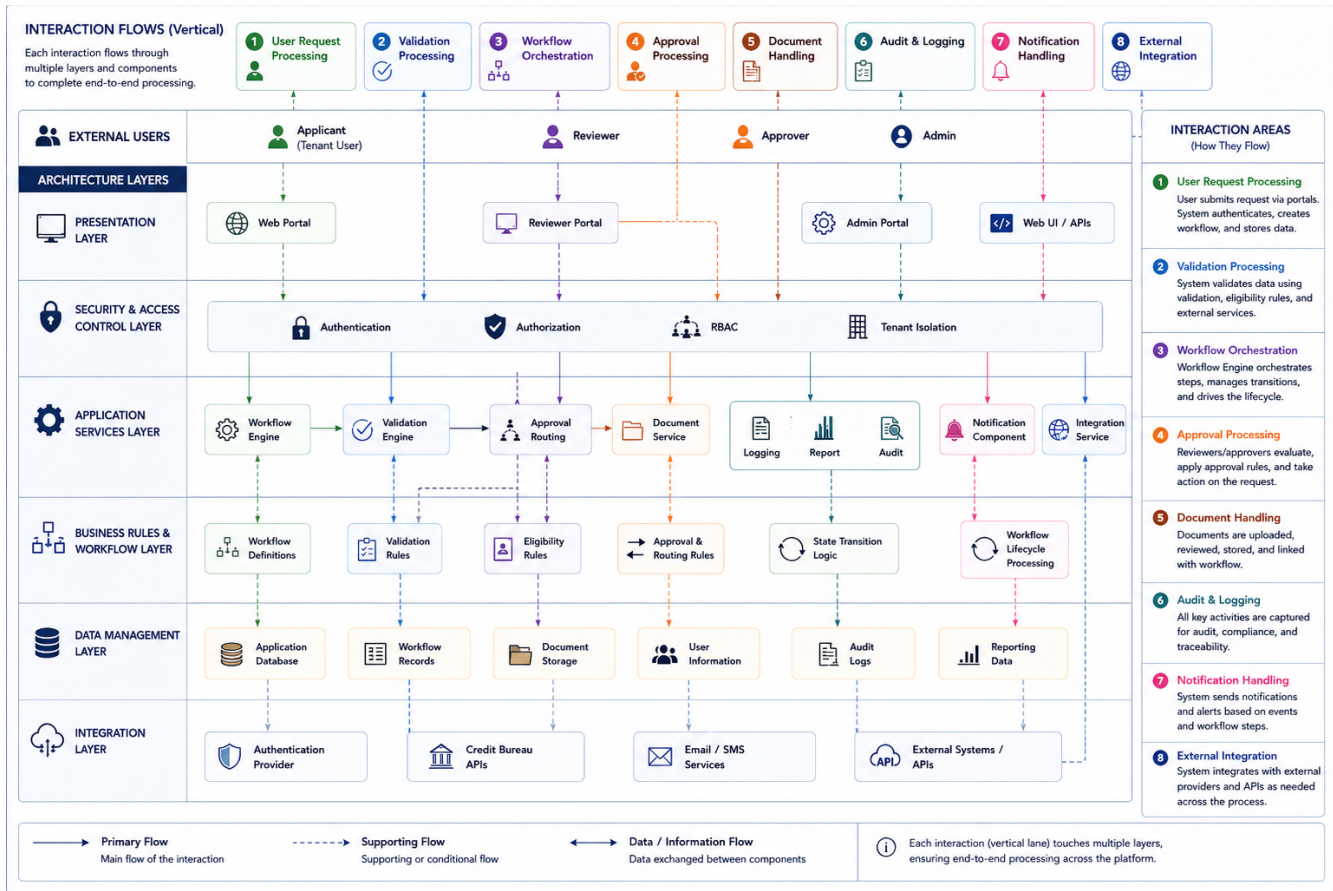


Figure 3 Component Interaction Overview

4. Workflow Architecture

This section describes how the **CardFlow Credit Origination Platform** orchestrates credit application, amendment, and reinstatement workflows through a shared lifecycle model. It explains how workflow services, validation logic, routing rules, status transitions, audit logging, and notifications work together to process requests from initiation to outcome.

4.1. Workflow Lifecycle Architecture

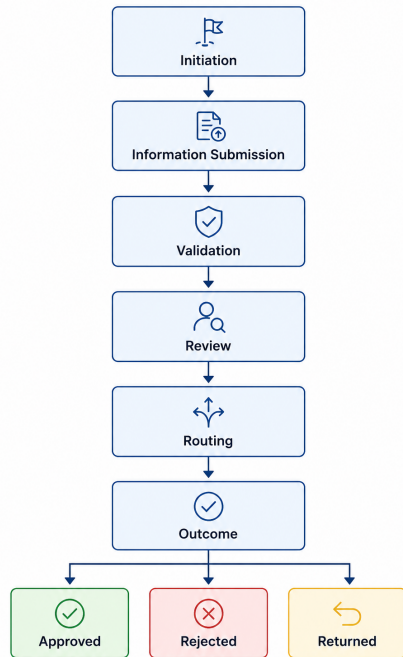


Figure 4 Workflow Lifecycle Architecture

The **Workflow Lifecycle Architecture** defines the common processing model used across all supported workflows. Each request follows a structured sequence of stages, allowing the platform to manage request progression consistently while applying workflow rules, validation checks, routing logic, and audit tracking.

4.2. Credit Application Processing Architecture

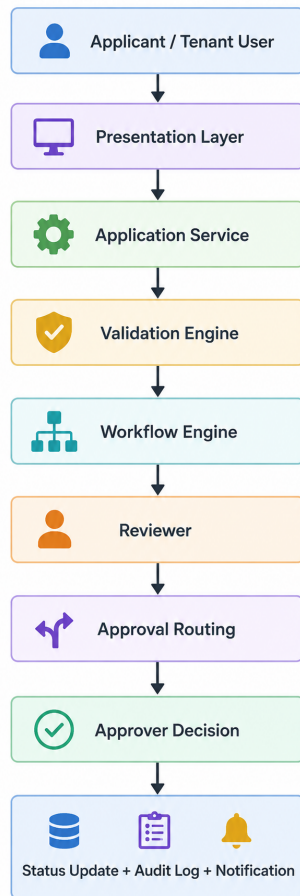


Figure 5 Credit Application Processing Architecture

The **Credit Application Processing Architecture** shows how the platform handles new credit applications from creation to final decision. The process begins when an applicant or tenant user initiates an application, continues through information submission and validation, and proceeds to review, approval routing, and outcome recording.

4.3. Amendment Processing Architecture

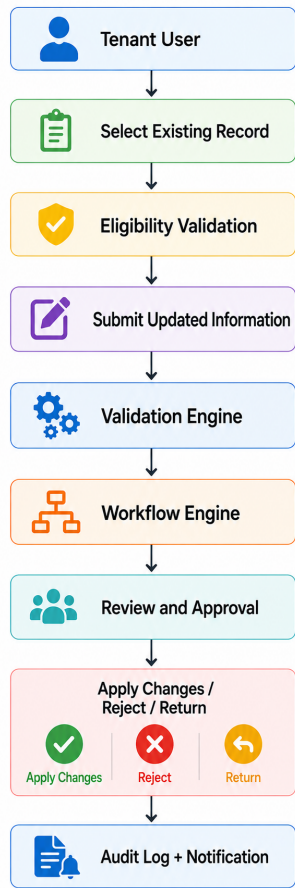


Figure 6 Amendment Processing Architecture

The **Amendment Processing Architecture** shows how the platform manages requests to update existing application or account information. This workflow includes an eligibility validation step before the request proceeds, ensuring that only eligible records can be amended.

4.4. Reinstatement Processing Architecture

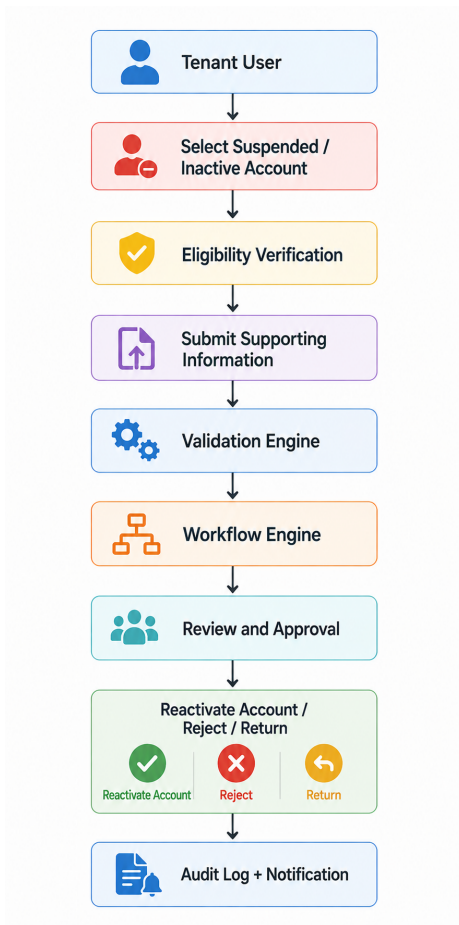


Figure 7 Reinstatement Processing Architecture

The **Reinstatement Processing Architecture** shows how the platform processes requests to reactivate suspended or inactive credit accounts. The workflow focuses on account status validation, eligibility checks, supporting documentation, review, approval, and final account reactivation or rejection.

5. Data Architecture

This section describes how the **CardFlow Credit Origination Platform** manages the key data objects required to support credit application, amendment, and reinstatement workflows. It covers the core business entities, their relationships, and the movement of data through validation, workflow processing, storage, audit logging, and reporting.

5.1. Core Business Entities

This subsection identifies the core business entities managed by the **CardFlow Credit Origination Platform** and provides a high-level description of the primary data objects that support workflow processing, document handling, audit tracking, user management, and tenant-specific configuration.

Table 5 Core Business Entities

Entity	Description
Applicant	Represents the customer or individual associated with a credit application, amendment, or reinstatement request.
Credit Application	Represents the main credit request submitted for validation, review, routing, and approval.
Workflow Instance	Represents the lifecycle record of a request as it moves through workflow stages and states.
Document	Represents uploaded supporting files linked to an application or request.
User	Represents platform users such as applicants, tenant users, reviewers, approvers, and administrators.
Audit Record	Represents logged user actions, system events, status changes, and approval decisions.
Tenant Configuration	Represents tenant-specific settings such as roles, approval rules, workflow configuration, and required document lists.
Notification Record	Represents system-generated notification events triggered by workflow actions or status changes.

5.2. Entity Relationship Diagram

This subsection presents a high-level relationship view of the core data entities managed by the platform.

Note:

- The diagram is conceptual and does not define database tables, fields, keys, or physical schema details.
- The **Entity Relationship Diagram (ERD)** uses simplified **Crow's Foot Notation** to represent relationships between the core business entities of the **CardFlow Credit Origination Platform**.

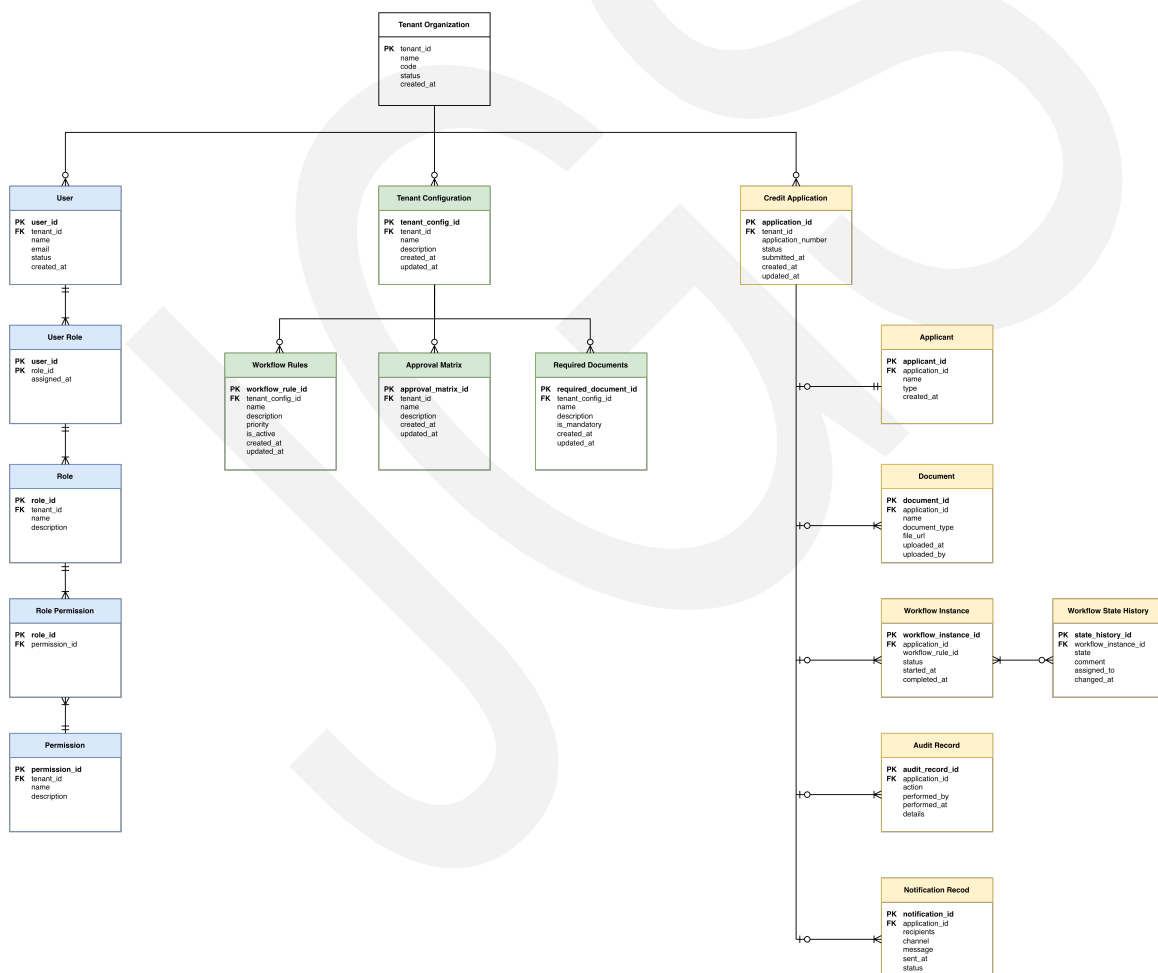


Figure 8 Entity Relationship Diagram

5.3. Data Flow Architecture

This subsection describes how data moves through the platform during workflow processing. User-provided information and supporting documents are captured through the presentation layer, validated by the system, processed through workflow services, stored in the data layer, and recorded for audit and reporting purposes.

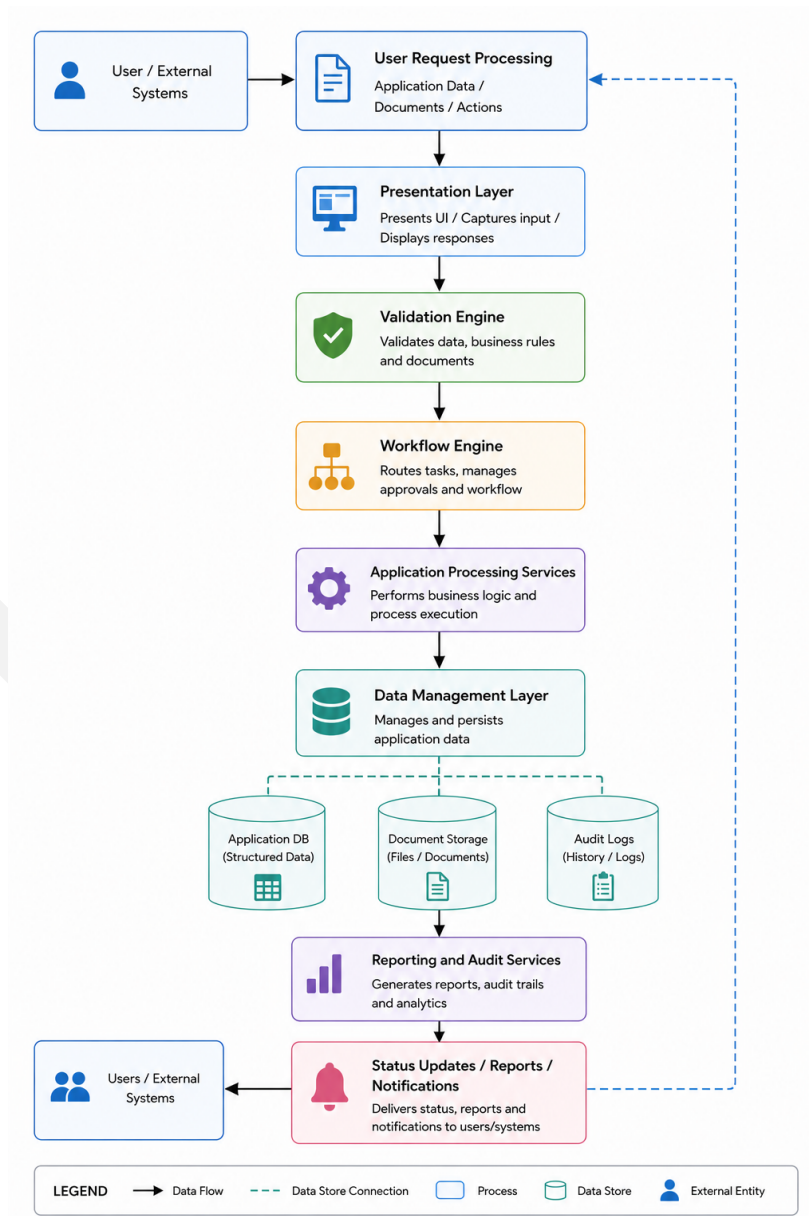


Figure 9 Data Flow Architecture

6. Security Architecture

This section describes the security architecture of the **CardFlow Credit Origination Platform**, including the authentication, access control, audit, traceability, and data protection mechanisms that safeguard workflow processing, enforce tenant isolation, and protect sensitive information within the multi-tenant SaaS environment.

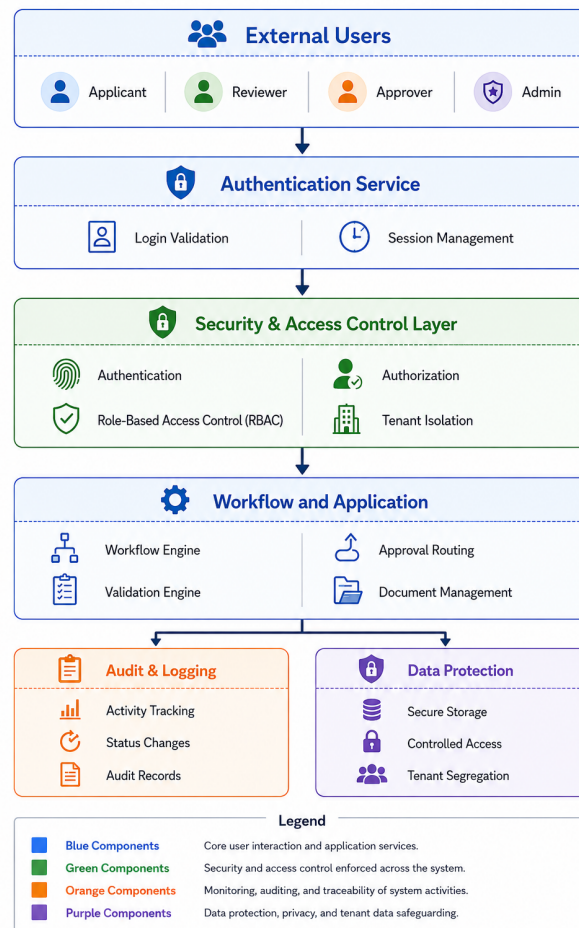


Figure 10 Security Architecture

6.1. Authentication and Access Control

This subsection describes the authentication and authorization mechanisms used by the **CardFlow Credit Origination Platform** to control access to system resources and workflow operations.

The platform enforces **role-based access control (RBAC)** to ensure that users can only access functions, workflow actions, records, and administrative capabilities permitted by their assigned roles and tenant-specific permissions.

The security and access control architecture supports:

- User authentication and session management
- Role-based access control (RBAC)
- Tenant-level data isolation
- Permission-based workflow access
- Controlled administrative access
- Secure credential validation
- Authorization enforcement across workflow stages

The platform supports multiple user roles, including applicants, tenant users, reviewers, approvers, and administrators, each with distinct operational permissions and workflow responsibilities.

6.2. Audit and Traceability

This subsection describes the audit and traceability mechanisms implemented within the **CardFlow Credit Origination Platform** to support workflow transparency, operational monitoring, compliance tracking, and historical record preservation.

The platform records workflow-related activities and system events throughout the request lifecycle, including:

- Request creation and submission
- Validation events
- Workflow state transitions
- Review and approval decisions
- User actions and updates
- Notification activities
- Administrative configuration changes

Audit records are stored and maintained to support:

- Workflow traceability

- Operational review
- Compliance monitoring
- Historical activity tracking
- Reporting and investigation activities

The audit architecture ensures that workflow actions and system-generated events can be traced throughout the lifecycle of a request across tenant environments.

6.3. Data Protection

This subsection describes the data protection mechanisms implemented within the **CardFlow Credit Origination Platform** to support secure storage, transmission, and handling of application and customer information.

The platform applies security controls and protection mechanisms designed to preserve the confidentiality, integrity, and controlled access of sensitive workflow and customer data within the multi-tenant SaaS environment.

Data protection measures include:

- Secure transmission of data between platform components
- Controlled access to sensitive records and documents
- Tenant-level data segregation
- Protection of uploaded documents and audit records
- Controlled access to workflow and reporting information
- Secure handling of authentication and session information
- Audit preservation and traceability controls

The platform architecture is designed to support secure operational processing while maintaining workflow accessibility, auditability, and tenant-specific access boundaries across the system.

7. Integration Architecture

This section describes how the **CardFlow Credit Origination Platform** integrates with external systems and third-party services to support authentication, workflow processing, notification delivery, and data exchange. The architecture ensures secure, controlled, and consistent interactions while supporting workflow execution, audit logging, and operational reliability across the platform.

7.1. External System Interfaces

This subsection identifies the external systems and services that may interact with the **CardFlow Credit Origination Platform** to support authentication, communication, data validation, and operational processing.

Table 6 External System Interfaces

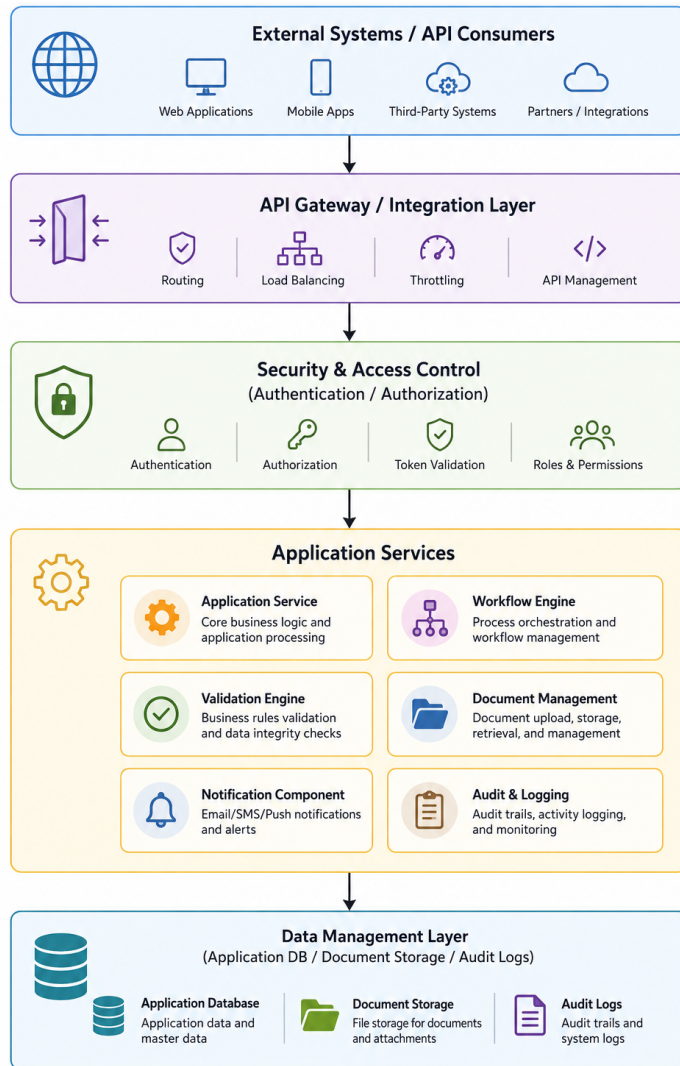
External System / Service	Purpose	Interaction Type
Authentication Provider	Supports secure user authentication and identity validation.	Authentication / authorization
Email / SMS Service	Delivers workflow-triggered notifications to applicants, reviewers, approvers, and administrators.	Notification delivery
Credit Bureau / External Credit System	Supports retrieval or validation of customer-related or credit-related information, where applicable.	Data lookup / validation
External API Consumers	Allows authorized external systems to create, retrieve, or update credit application data through secure APIs.	API integration
Reporting / Analytics Tools	May consume application, workflow, or audit data for operational reporting and monitoring.	Data export / reporting

7.2. API Architecture

This subsection describes how API-based interactions are structured within the **CardFlow Credit Origination Platform**. APIs allow internal and external systems to exchange data securely with the platform while supporting workflow-related operations such as application creation, data retrieval, status updates, and external validation.

The API architecture supports:

- Secure API access through authentication and authorization.
- Request validation before processing.
- Controlled access to application and workflow data.
- Integration with workflow, validation, document, audit, and notification services.
- Consistent response handling for successful and failed API requests.



When an external system sends a request to create a credit application:

1. The request enters through the **API Gateway / Integration Layer**.
2. The **Security & Access Control** layer checks authentication and permissions.
3. The request is passed to the **Application Service**.
4. The **Validation Engine** checks required data.
5. The **Workflow Engine** creates the workflow record.
6. The **Data Management Layer** stores the application.
7. The **Audit & Logging** component records the event.

Figure 11 API Architecture

x

Table 7 API Architecture

Diagram Area	Descriptions
External Systems / API Consumers	Third-party systems or partner applications that need to exchange data with CardFlow.
Integration Layer/ API Gateway	Entry point for API requests. It receives requests, routes them, and standardizes communication.

Diagram Area	Descriptions
Security & Access Control	Verifies whether the request is authenticated and authorized before allowing access.
Application Services	Handles the actual business operation, such as creating an application, validating data, updating workflow status, sending notifications, or writing audit logs.
Data Management Layer	Stores or retrieves application records, documents, workflow data, and audit logs.

7.3. Notification Architecture

This subsection describes how the **CardFlow Credit Origination Platform** generates and delivers workflow-triggered notifications to keep users informed of request status changes and required actions. The architecture supports timely communication, event-driven notification delivery, and audit logging for traceability.

Table 8 Notification Triggers

Trigger Event	Target User
Request submitted	Reviewer / System users
Request assigned for review	Reviewer
Request assigned for approval	Approver
Request returned for correction	Applicant / Tenant User
Request approved	Applicant / Tenant User
Request rejected	Applicant / Tenant User
Workflow exception occurs	Relevant operational user / Admin

The workflow engine triggers a notification event when a request reaches a defined workflow point.

The **Notification Component** evaluates the **event type**, **user role**, and **workflow stage** to determine who should receive the notification and through which channel.

After delivery, the platform records the notification event for audit and reference purposes.

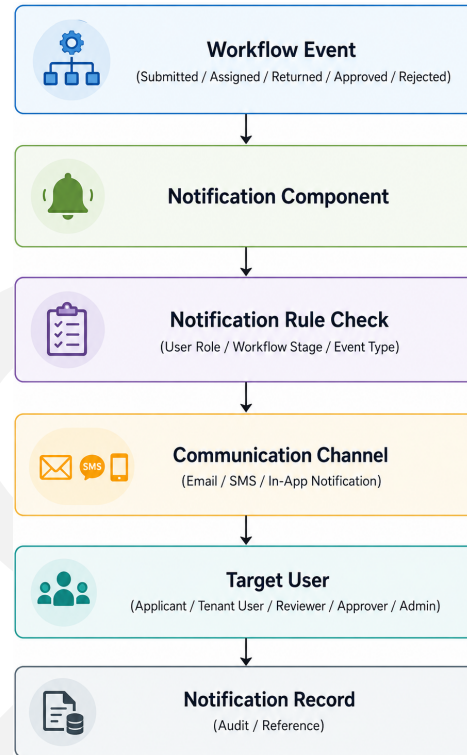


Figure 12 Notification Architectures

8. Deployment Architecture

This section describes the high-level deployment structure of the **CardFlow Credit Origination Platform**. It shows how users access the platform, how application services are hosted, and how supporting components such as databases, document storage, audit logs, and external integrations are deployed within a cloud-based SaaS environment.

8.1. Environment Overview

The **CardFlow Credit Origination Platform** is deployed as a cloud-hosted, multi-tenant SaaS application accessible through supported web browsers. Users interact with the platform through web-based interfaces, while backend services process workflow actions, validation requests, approvals, notifications, reporting, and audit events.

Table 9 Environment Overview

Environment Area	Description
User Access	Users access the platform through a browser-based interface.
Web Layer	Hosts the web application and user-facing interface components.
Application Layer	Hosts backend services such as workflow, validation, approval routing, reporting, and notifications.
Data Layer	Stores application records, workflow data, documents, audit logs, and configuration data.
Integration Layer	Supports communication with authentication services, notification providers, and external systems.
Cloud Environment	Provides hosting, scalability, availability, monitoring, and infrastructure support.

8.2. Deployment Diagram

This subsection presents the high-level deployment structure of the **CardFlow Credit Origination Platform**, illustrating how users, application services, data storage components, and external systems are distributed and connected within the cloud-based SaaS environment.

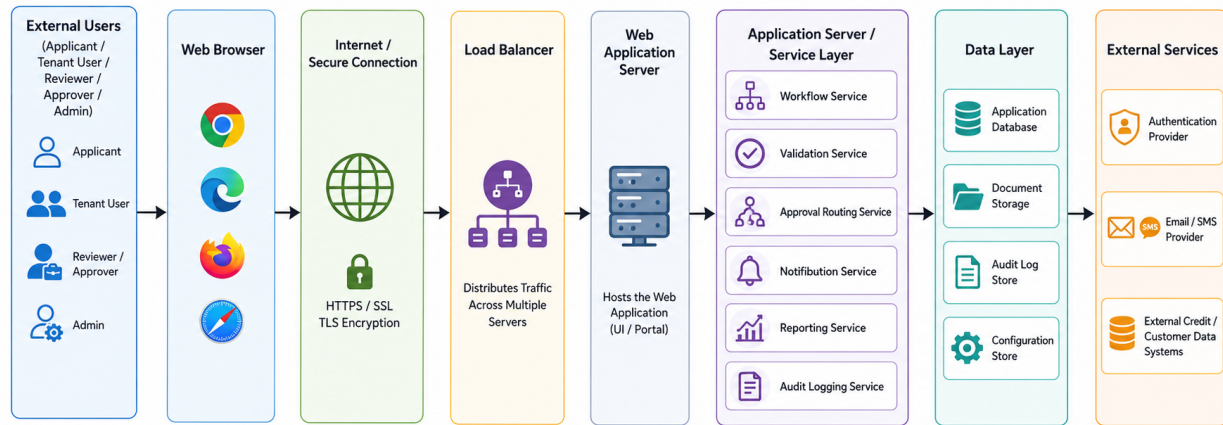


Figure 13 Deployment Diagram

8.3. Scalability and Availability Considerations

The deployment architecture supports scalability and availability by separating user access, web delivery, application services, data storage, and external integrations into distinct deployment areas.

Table 10 Scalability and Availability Considerations

Consideration	Description
Scalability	Application services can scale to support increasing users, tenants, and workflow transactions.
Availability	Load balancing and cloud-hosted deployment help maintain access to platform services.
Reliability	Separation of application services and data storage supports stable processing and recovery.
Tenant Growth	The platform supports additional tenant organizations without changing the core application model.
Service Isolation	Core services such as workflow, validation, reporting, and notification processing are logically separated.
Operational Continuity	Audit logs, workflow records, and stored documents support continuity and traceability during operational review or recovery.

9. Architecture Considerations

This section describes the key architectural considerations applied during the design of the **CardFlow Credit Origination Platform** to support scalability, maintainability, security, auditability, and tenant isolation within the multi-tenant SaaS environment.

The platform architecture is designed to support the following while maintaining separation between tenant environments and supporting enterprise-level operational requirements:

- Workflow-driven processing
- Configurable business rules
- Secure operational handling
- Sustainable system growth

9.1. Scalability

The platform architecture supports scalability by separating presentation, application processing, workflow orchestration, data management, and integration services into distinct architectural layers. This separation allows the platform to support increasing users, tenants, workflow transactions, and operational workloads without significantly impacting overall system behavior.

Scalability considerations include:

- Support for multiple tenant organizations
- Separation of processing services and data management
- Expandable workflow and approval processing
- Support for increasing workflow volumes and concurrent users
- Flexible integration capabilities for external systems and services

9.2. Maintainability

The platform architecture is designed to support maintainability by organizing system capabilities into modular functional components and clearly separated architectural layers. This structure allows workflow logic, validation rules, integrations, reporting services, and notification behavior to be updated or extended with minimal impact on other platform components.

Maintainability considerations include:

- Modular service-oriented architecture
- Separation of business rules and workflow logic
- Configurable workflow and approval processing
- Centralized validation and notification handling
- Structured audit and reporting capabilities

9.3. Security

The platform architecture incorporates security controls to support secure access management, tenant-level isolation, controlled workflow access, and protection of sensitive application and customer information.

Security considerations include:

- Authentication and authorization enforcement
- Role-based access control (RBAC)
- Tenant-specific access boundaries
- Controlled access to workflow and reporting functions
- Secure handling of documents and audit records
- Protection of workflow and application data

9.4. Auditability

The platform architecture supports auditability by recording workflow activities, approval actions, validation events, state transitions, and system-generated operations throughout the request lifecycle.

Auditability considerations include:

- Workflow traceability across lifecycle stages
- Recording of user and system actions
- Audit preservation for operational review
- Tracking of workflow state transitions
- Notification and approval event logging
- Support for reporting and compliance review activities

9.5. Tenant Isolation

The platform architecture supports tenant isolation by logically separating tenant-specific users, workflow configurations, approval rules, documents, and operational data within the shared SaaS environment.

Tenant isolation considerations include:

- Tenant-specific access control
- Logical separation of workflow and application data
- Tenant-level configuration management
- Segregated reporting and operational visibility
- Controlled access to tenant-owned records and documents
- Secure handling of tenant-specific workflow operations

10. Assumptions and Constraints

This section identifies the architectural assumptions and operational constraints considered during the design of the **CardFlow Credit Origination Platform**. These assumptions and constraints help define the expected operating conditions, integration dependencies, workflow behavior, and architectural boundaries of the platform in the multi-tenant SaaS environment.

10.1. Assumptions

The following assumptions were considered during the architectural design of the platform:

Table 11 Assumptions

Assumption	Description
Cloud-Based Deployment	The platform is deployed in a cloud-hosted SaaS environment.
Tenant-Based Operations	Multiple tenant organizations share the platform while maintaining logical data and access isolation.
Role-Based Access Control	Users are assigned roles that define workflow access and operational permissions.
Workflow-Driven Processing	Credit application, amendment, and reinstatement requests follow defined workflow lifecycle stages.
External Service Availability	External authentication, notification, and integration services are assumed to be operational and accessible when required.
Browser-Based Access	Users access the platform through supported web browsers and secured network connections.
Configurable Workflow Rules	Workflow routing, approval conditions, and validation rules are configurable at the tenant or platform level.
Audit Preservation	Workflow actions and system-generated events are recorded and retained for operational traceability.

10.2. Constraints

The following constraints apply to the architectural design and operational behavior of the platform:

Table 12 Constraints

Constraint	Description
Shared SaaS Infrastructure	Multiple tenant organizations operate in a shared platform environment.
Tenant Isolation Requirements	Tenant data and workflow operations must remain logically segregated.
Dependency on External Services	Authentication providers, notification services, and external systems may impact workflow processing if unavailable.
Workflow Dependency	Request processing is dependent on configured workflow definitions, routing rules, and approval structures.
Role-Based Operational Limits	User actions are restricted based on assigned roles and permissions.
Document Storage Dependency	Workflow processing may require uploaded supporting documents before progression or approval.
Audit and Logging Requirements	Workflow actions and operational events must be recorded for traceability and reporting purposes.
Integration Dependency	API-based operations depend on availability and compatibility of external integration endpoints.